Shyam Patel

Data Scientist

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Summary

- With over 2.5+ years of experience as a Data Engineer and AI/ML Engineer, I have worked on data migrations, AI/ML projects, and Generative AI applications.
- Recently, I have been leading an AI team, driving innovation in LLM development, RAG pipelines, and AI-driven automation.
- My expertise spans Machine Learning Algorithms, Deep Learning Architectures, AWS technologies, Generative AI, and NLP, along with Snowflake database migration, and Python programming.
- Beyond technical work, I am passionate about knowledge sharing. I write blogs on Medium, where I break down AI advancements, LLM best practices, and real-world AI applications, making complex concepts accessible to a wider audience

Experience

Kenexai Ahmedabad

Data Scientist (Dec-2022 – Present)

- Developed AI-powered solutions, leveraging Machine Learning models, LLMs, AWS and advanced data processing techniques to drive innovation and automation.
- Delivered scalable and data-driven AI solutions, integrating Generative AI, NLP, and predictive analytics to meet complex business requirements.
- Spearheaded a key role in the Finance Project and Home Rental Analysis Project, collaborating with the Data Engineering team to successfully migrate data across platforms using AWS, Oracle, Snowflake, and Informatica Cloud.
- Optimized data workflows and enhanced data quality, ensuring seamless data migration, efficient processing, and cost optimization.

MEDIUM Blogger– (Sept-2023 – Present)

- Passionately create and publish insightful, beginner-friendly blogs on Medium.com, aimed at demystifying complex data science concepts for a wider audience.
- Utilize relatable real-life examples to simplify intricate data science topics, attracting an expanding readership and receiving positive feedback from the data science community.

Panth Softech - Software Development Company Ahmedabad:

Data Scientist (Intern) - (May 2022 – July 2022)

- Collaborated with a team of data scientists to develop an innovative image-based deep learning model for automatically extracting temperature panel digits from milk container images.
- Conducted image annotations, participated in model building part.

Skills

Category	Technologies & Tools
Cloud Platforms	AWS (S3, Glue, Lambda, SES, Bedrock, Neptune, RDS, OpenSearch, Sagemaker)
Programming	Python (ML Pipelines, AI Models, LLMs), SQL
Generative AI	RAG, Langchain, LangGraph, LlamaIndex, OpenAI, LLM as Judge, CrewAI, Agents
Machine Learning & Deep Learning	ML Algorithms, Neural Network, Transformer, CNN, LSTM
NLP	Named Entity Recognition (NER) models, Text Classification, Text Generation, Metadata Extraction, TTS, STT
Data Platforms & Visualization	Snowflake, Tableau
Version Control	Git, Github
Project Management	JIRA, Confluence
Technical Writing	AI/ML Blog Writing (Medium) on LLMs, AI Advancements, and Best Practices
Soft Skills	Quick Learner, Creativity, Problem Solving

Projects

AI powered Interview System

Technologies Used: Python, Crew AI, OpenAI, Sql-lite
Project Description: Using AI-driven questioning, resume analysis, and adaptive learning, it enables a fair, scalable, and data-driven hiring process.
Roles and Responsibilities:

- Developed an AI-powered interview bot that analyzes user resumes to extract key skills, technologies, and project descriptions.
- Integrated a large language model (LLM) to understand the context of user-provided resumes and generate interview questions tailored to the applicant's expertise.
- Designed a domain-specific decision-making system where the AI selects appropriate panelists for the interview (e.g., for a Data Engineer, panels include Python, SQL, and Cloud).
- Implemented evaluation steps to assess candidate responses, scoring them on relevance, accuracy, and depth of Understanding.
- Enhanced interactivity by enabling the AI to ask counter-questions on specific topics if clarification or deeper insight is required based on user responses.
- Streamlined the first round of interviews by automating the generation of domain-relevant questions and dynamically adjusting the flow of the interview.
- Improved recruitment efficiency by automating the initial screening process, significantly reducing manual effort.

AWS-Powered AI Chatbot

Technologies Used: Python, Langchain, LlamaIndex, Bedrock (Claude), PostgreSQL, AWS (RDS, Neptune, S3, OpenSearch, EC2)

Project Description: Developed an AI-driven chatbot using Retrieval-Augmented Generation (RAG) to retrieve answers from structured and unstructured data sources. The chatbot is built on a full AWS stack, ensuring seamless data retrieval and intelligent response generation.

Roles and Responsibilities:

- Designed and implemented a chatbot leveraging AWS Bedrock (Claude) for LLM-powered natural language interactions.
- Managed structured data in RDS, transforming it into a graph-based model in Neptune for advanced data relationships and querying.
- Constructed SQL queries on Neptune to extract insights from RDS efficiently.
- Handled unstructured data in S3, utilizing OpenSearch for vector storage and intelligent search.
- Deployed a Django-based chatbot application on EC2, integrating all AWS components for a seamless experience.
- Optimized query performance and knowledge retrieval, ensuring context-aware and real-time responses.

DB -Chatbot

Technologies Used: Python, Langchain, Llamaindex, OpenAi, Postgresql

Project Description: Developed a dialogue-based chatbot using Retrieval-Augmented Generation (RAG) to retrieve answers from database files (DB) and PDF documents.

Roles and Responsibilities:

- Designed and developed a dialogue-based chatbot with Retrieval-Augmented Generation (RAG) for enhanced contextual understanding
- Implemented RAG using Llama Parser to improve document retrieval and response accuracy.
- Optimized embedding storage and retrieval for efficient query handling.
- Integrated vector databases for efficient knowledge retrieval and response generation.
- Enhanced chatbot adaptability with real-time query resolution and context-aware responses.

Automatic Trip Generation

Technologies Used: Python, AI/ML Algorithms, Maps API, Database Integration

Project Description: Developed a custom logistic system to automate delivery trip planning, optimizing vehicle utilization, reducing delivery time, and minimizing costs through intelligent shipment grouping and route optimization.

Roles and Responsibilities:

- Designed Automated Trip Creation to minimize manual effort by dynamically grouping shipments.
- Implemented Route Optimization using Maps API for real-time traffic-aware routing.
- Developed Dynamic Adjustments to integrate new shipments into existing trips with minimal disruption.
- Built Auto-Adjustment & Merging Mechanisms to optimize vehicle utilization by combining underutilized trips.
- Enabled Manual Adjustments for fulfillment center users to fine-tune trip allocations.
- Ensured Scalability by supporting multiple vehicle types and handling fluctuating order volumes.
- Improved Delivery Timeliness by sequencing shipments efficiently within designated time slots.
  Enhanced Operational Efficiency by reducing errors and automating logistics processes.

AI-Based Email Folder Recommendation System

Technologies Used: Python, Sentence Transformer, Pinecone Vector Database, Named Entity Recognition (NER)

Project Description: Developed an AI-powered system that suggests appropriate folder names for incoming emails using unsupervised learning and similarity-based search, improving email organization efficiency.

Roles and Responsibilities:

- Implemented Text Processing techniques, combining email subject and body while applying preprocessing steps.
- Designed Vector Generation using Sentence Transformer embeddings for efficient text representation.
- Integrated Pinecone Vector Database to store and retrieve high-dimensional embeddings with metadata filtering.

- Applied Named Entity Recognition (NER) for extracting relevant organization-related keywords.
- Developed an Unsupervised Learning Model leveraging similarity search for email classification.
- Achieved 93% accuracy in test cases while ensuring adaptability to new folder structures.

Object Tracking - Flappy Bird Game Using Computer Vision

Technologies Used: Python, OpenCV, Pygame, Computer Vision
Project Description: Developed a Flappy Bird game with an innovative object tracking feature using computer vision techniques. The game utilizes a webcam to track a specific object color, allowing players to control the bird's movement in real time. If the object color is not detected, the game ends.
Roles and Responsibilities:

- Developed the Flappy Bird game mechanics using Python and the Pygame library for GUI development.
- Implemented real-time object tracking using OpenCV, enabling players to control the bird via webcam movement.
- Designed computer vision algorithms to detect and track specific object colors within the video stream.
- Optimized tracking performance to ensure smooth gameplay and accurate object recognition.
- Enhanced user interaction by integrating a dynamic real-time feedback system based on object visibility.

AI-Powered Smart Dashboard Generator and Enhancer

Technologies Used: Tableau, Python, OpenAI API, Pandas, Scikit-learn, LangChain
Project Description: Developed an AI-driven system that automates chart recommendations based on raw datasets and analyzes existing dashboards to suggest improvements for better business insights.
Roles and Responsibilities:

- Implemented genAI models to analyze datasets and recommend the most effective chart types (bar, line, scatter, heatmaps, etc.).
- Developed a reverse analysis engine that evaluates existing dashboards, highlights missing insights, and suggests data-driven enhancements.
- Integrated natural language querying, allowing users to input business questions and receive AI-generated charts with insights.
- Used anomaly detection to identify critical data patterns and suggest visual representations to highlight them.
- Automated KPI tracking by recommending relevant metrics and visuals based on historical data trends.
- Enhanced business decision-making by streamlining data-to-dashboard workflows with AI-powered suggestions.

Container Damage Detection

Technologies Used: OpenCV, Machine Learning, Deep Learning, Flask
Project Description: Developed an AI-powered computer vision system for real-time detection and classification of container damages such as dents, cracks, and corrosion, ensuring cargo safety and operational efficiency.
Roles and Responsibilities:

- Developed a user-friendly interface using Flask for real-time container damage detection.
- Implemented computer vision techniques to detect and classify damages like dents, cracks, and corrosion.
- Built deep learning models using OpenCV to analyze images and identify defects in shipping containers.
- Designed and integrated a system to visualize detected defects and provide detailed location information.
- Optimized the system for real-time analysis to ensure operational efficiency and cargo safety.
- Worked closely with stakeholders to gather requirements and ensure the system met their needs.
- Deployed the system in a real-world environment, enabling real-time detection of container defects.

Speech-to-Text and Text Summarization Project

Technologies Used: Speech Recognition, NLP, Machine Learning
Project Description: Developed an AI-driven speech-to-text and summarization system that converts spoken language into text and generates concise summaries, enhancing content accessibility and productivity.
Roles and Responsibilities:

- Developed speech recognition algorithms to convert spoken language into text, handling challenges like accents and background noise.
- Implemented NLP models to summarize long-form content into concise summaries.
- Integrated speech recognition and NLP technologies to create an end-to-end solution for speech-to-text conversion and summarization.
- Optimized the system to handle different languages and dialects for broader applicability.
- Collaborated with other engineers to fine-tune models and improve the system's overall performance.
- Tested the system with various datasets to ensure robustness and accuracy.

Finance Project - Snowflake Data Migration

Technologies Used: Snowflake, Oracle, SQL, Informatica Cloud, Python

Project Description: Led the migration of financial data from Oracle to Snowflake, enabling a scalable, cloud-based data storage solution. The project involved handling daily and hourly data influx, integrating multiple data sources, and automating data pipelines to ensure seamless migration and efficient processing.
Roles and Responsibilities:

- Migrated financial data from Oracle databases to Snowflake, ensuring data integrity and consistency.
- Collaborated with clients to transition data storage to Snowflake, managing real-time and historical data.
- Worked closely with business users and analysts for requirements gathering and analysis.
- Designed and developed high-level and low-level data architecture based on business needs.
- Implemented Snowflake scripts, stored procedures, and pipelines to automate and optimize data migration.
- Integrated multiple data sources into the staging area for data cleansing and transformation.
- Created and managed Snowflake objects, including databases, schemas, external tables, streams, stored procedures, and tasks, to support ETL workflows.

Achievements and Certifications

NPTEL Certification: Successfully completed the NPTEL Exam with a consolidated score of 72% in Python for DataScience
Global Representation at GITEX 2024, Dubai: Selected by my company to attend the world's biggest AI event, GITEX 2024 in Dubai, representing the organization and showcasing its innovations.
Coursera Certification: successfully completed the 6 Course Program by Google, which is Google IT Automation with Python.
SnowPro Core Certification Exam (COF-C02): Achieved a score of 700/1000 in the Snowflake Certification Exam.
In addition, I have participated in various hackathons and completed multiple courses on platforms such as Coursera, Great Learning, and Udemy. I also actively solve problems on platforms like HackerRank, LeetCode, and GeeksforGeeks.